

INTERNSHIP REPORT

ON

FIBER OPTIC COMMUNICATIONS, FTTH NETWORK ARCHITECTURE AND ELECTRICAL HOME APPLIANCES MAINTENANCE



Submitted by:

DHIYA FATHIMA MP
Department of Physics
Sullamussalam Science College, Areekode

Under the Faculty Guidance of:
Dr. Mohemmed Shanid N A
Department of Physics

CERTIFICATE

This is to certify that the internship report entitled “Fiber Optic Communications, FTTH Network Architecture and Electrical Home Appliances Maintenance” is a bona fide record of the industrial training carried out by **DHIYA FATHIMA MP** under my academic supervision and guidance, in corporate collaboration with Bfone Fibernet, Edavannappara, in partial fulfillment of the requirements for the

Dr. Mohemmed Shanid N A

award of the Degree of Bachelor of Science in Physics. Head of the Department

Department of Physics

External Examiner

Internal Examiner

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Internship Activity Schedule

The day-wise internship activities and topics covered during the training period are summarized in the table below:

Day	Date	Topics / Activities Covered
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)
Day 2	22.05.2026	Splicing Practical Section
Day 3	23.05.2026	Repairing of the Ceiling Fan
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange
Day 6	08.06.2026	Power Calculation Theory Class
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker

Table 1: Day-wise Internship Activity Log

Part I

Fiber Optic Communications and FTTH Network Architecture

Part Introduction: Fiber Optic Infrastructure

The modern telecommunication landscape is fundamentally powered by high-speed data transmission systems that rely on optical physics. Transitioning away from traditional legacy copper wire infrastructures, optical fibers utilize light pulses to transmit vast amounts of data over immense distances with minimal attenuation. This part of the internship report establishes a detailed technical study of modern Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) network architectures, directly informed by industrial training with Bfone Fibernet. Our main goal in this part is to trace the complete pathway of the optical signal from the central exchange to residential installations. The chapters in this section study Total Internal Reflection (TIR) guiding light in glass cores, multi-core cable design, and the calculations needed for fiber link budgets. Additionally, this part covers real-world field operations like fusion splicing, fiber color coding, and diagnosing link faults in active environments. The module wraps up with observations made during the field visit to the local BSNL Exchange.

Chapter 1

Introduction to Optical Fiber Communication

1.1 Overview

Optical Fiber Communication (OFC) is a method of transmitting information from one place to another by sending pulses of light through an optical fiber. The light forms an electromagnetic carrier wave that is modulated to carry information.

1.2 The Basic Fiber Optic Link

A typical optical fiber communication link consists of several key components that facilitate the transmission of data:

- Encoder/Source: Takes input data and translates it into digital electrical pulses.
- Transmitter (E/O): Uses a semiconductor LED or laser diode to convert electrical pulses into light signals.
- Optical Fiber: The silica glass or plastic medium that guides the light pulse over a distance.
- Receiver (O/E): Employs a photodiode (PIN or APD) to detect the light signal and convert it back to electrical form.
- Decoder: Reconstructs the original digital data from the electrical pulses.

1.3 Advantages of Fiber Optics

- Non-conductive: Made of glass/plastic, eliminating electrical spark risks and grounding problems.
- Electromagnetic Immunity: Unaffected by electromagnetic interference (EMI), radio frequency interference, or lightning strikes.
- Large Bandwidth: Capable of supporting incredibly high data rates and channel capacities.
- Low Loss: Extremely low signal attenuation, allowing long spans without repeaters.
- Small and Lightweight: Thin profile and lightweight nature simplify installation in conduits and aerial runs.

Chapter 2

Principle and Cable Structure

2.1 Principle of Operation: Total Internal Reflection

Light travels through the fiber core based on the principle of Total Internal Reflection (TIR). For TIR to occur, two conditions must be satisfied:

- The core material must have a higher refractive index (n_1) than the surrounding cladding (n_2).
- The light must strike the core-cladding boundary at an angle of incidence exceeding the critical angle (θ_c).

The critical angle is calculated as:

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1)$$

2.2 Cable Anatomy

A standard OFC cable consists of several protective and structural layers:

- Core: The central glass or plastic channel through which the light pulses travel.
- Cladding: The surrounding layer of glass with a lower refractive index that keeps the light trapped in the core.
- Buffer Coating: A plastic layer surrounding the cladding to shield it from moisture and mechanical stress.
- Strength Member: High-tensile aramid yarn (Kevlar) or steel wire to absorb tension during pulling.
- Outer Jacket: The tough plastic outer sheath (PVC or PE) protecting the cable from environmental factors.

2.3 Cable Types

- Tight-Buffer Cable: Features thick buffer coating directly on the fiber, ideal for indoor routing and patch cords.
- Loose-Tube Cable: Fibers lie loose inside gel-filled tubes, providing excellent protection against water and temperature shifts in outdoor ducts.
- Ribbon Cable: Multiple fibers aligned in flat rows, maximizing fiber density in tight spaces.

Chapter 3

Link Budgeting, Splicing, and Mathematical Calculations

3.1 OFC Splicing

Splices are permanent connections between two fibers. The main methods are:

- Fusion Splicing: Fusing the two glass ends together using a high-voltage electric arc.
- Mechanical Splicing: Aligning fiber cores within a precision alignment fixture and securing them.

3.2 Logarithmic Conversion Mechanics

Optical power values are measured using the logarithmic decibel-milliwatt (dBm) scale to make it easy to calculate losses across long fiber links:

$$P_{\text{dBm}} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right) \quad (3.1)$$

3.3 Splitter Loss Calculations

The theoretical insertion loss of an optical splitter scales based on the number of output ports (N):

$$\text{Loss}_{\text{ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

To perform reliable link calculations, the engineering team uses the following standard attenuation parameters:

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

Table 3.1: Standard Loss Parameters Used in FTTH Networks

3.5 Mathematical Verification: Power Link Budget Case Study

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

3.5.1 Case Study Parameter Blueprint

Consider a regional fiber run managed by Bfone Fibernet spanning a total distance of 7 km. The link includes 4 fusion splices, 2 SC/UPC connector pairs, a 1 × 16 splitter, and a 1 × 8 cascading splitter module. The total link budget metrics break down into the following distinct attenuation values:

- Fiber line loss: 7 km of fiber at 0.40 dB/km results in a loss of 2.8 dB
- First splitter stage (1 × 16) loss: 13.0 dB
- Second splitter stage (1 × 8) loss: 10.0 dB
- Fusion splice losses: 4 splices at 0.10 dB each equals 0.4 dB
- Connector losses: 2 SC/UPC pairs at 0.35 dB each equals 0.7 dB

Summing these individual loss values yields the total attenuation for the link:

$$\begin{aligned} \text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= 26.9 \text{ dB} \end{aligned}$$

3.5.2 Link Status Analysis

Standard ONT modems require an input signal level within an optical power range between -10 dBm and -28 dBm to operate reliably: Receiver Sensitivity Threshold $\leq$ -28 dBm

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

Because the calculated total loss for this link is 26.9 dB, a typical transmitter power output of 0 dBm ensures that the received power level stays above the receiver's minimum sensitivity threshold (-26.9 dBm $\geq$ -28 dBm). This mathematical verification confirms that the fiber link will function correctly and maintain high signal reliability.

Chapter 4

Optical Fiber Splicing: Detailed Analysis

4.1 What is Splicing?

Splicing is the process of joining two optical fiber cores permanently to create a continuous optical path. Unlike connectors, which are designed for frequent re-mating, splices are meant to be enduring and stable joints. The primary goal is to minimize optical loss and prevent back-reflection.

4.2 Types of Splicing

The main methods used in the industry are:

4.2.1 Fusion Splicing

This is the most reliable and industry-standard method. It involves using an electric arc to melt and fuse two fiber ends together. Loss: Typically 0.01 to 0.05 dB. Usage: Backbone networks, long-haul links, and FTTH.

4.2.2 Mechanical Splicing

This method uses a mechanical fixture to align the fiber ends and an index-matching gel to facilitate light transmission between the cores. Loss: Typically 0.2 to 0.5 dB. Usage: Emergency repairs, temporary setups, and environments where fusion equipment is not available.

4.3 Importance and Significance

- Link Extension: Joins shorter fiber spools to build long-distance network runs.
- Signal Integrity: Assures optimal light propagation and very low insertion loss.
- Back-reflection Control: Limits return loss (reflections) to protect high-power laser transmitters.
- Durability: Creates robust, long-lasting joints that resist temperature and physical strain.

4.4 Step-by-Step Fusion Splicing Process

1. Stripping: Stripping away the 250 μm polymer buffer coating using precision fiber strippers to expose bare glass.

2. **Cleaning:** Cleaning the bare glass surface with 99% pure Isopropyl Alcohol (IPA) and lint-free wipes.
3. **Cleaving:** Cutting the fiber using a precision cleaver to achieve a flat 90-degree angle (less than 0.5 degrees deviation).
4. **Fusion:** Aligning the fibers automatically using PAS and applying a localized electric arc to fuse the glass at 2000°C.
5. **Protection:** Sliding a heat-shrink protector sleeve over the splice point and baking it in the splicer's oven to seal it.

4.5 Advantages

- Provides the absolute lowest signal attenuation (≤ 0.05 dB).
- Forms a highly stable, permanent glass-to-glass bond.
- Unbothered by mechanical vibrations and thermal expansion.
- Very compact, fitting easily into splice trays and closures.

4.6 Future Scope

- **Automation:** Introduction of automated splicing machines and robotics for harsh environments.
- **Multi-core Fiber Splicing:** Upgrading equipment to splice multi-core fibers that support higher spatial bandwidth.
- **Enhanced Precision:** Incorporating nanometer-level alignment to reach near-zero signal loss in critical paths.

4.7 Conclusion

Optical fiber splicing is the foundation of modern high-speed communication. By ensuring a seamless and permanent joint, it guarantees that data travels across vast distances with negligible loss. While fusion splicing is the preferred method for permanent infrastructure, mechanical splicing remains a vital tool for rapid, field-based repairs.

Chapter 5

Introduction to FTTH & Passive Optical Networks (PON)

5.1 Overview of FTTH

Fiber-to-the-Home (FTTH) represents the ultimate evolutionary step in broadband access networks, replacing traditional copper-based infrastructures (like DSL or Coaxial cables) entirely with optical fiber lines directly to individual subscriber premises. Most modern FTTH networks utilize PON (Passive Optical Network) technology. The term “passive” is critical: it means the outer distribution network requires no electrical power between the provider’s central exchange and the customer’s home. Signals are routed and split using optical principles alone, drastically lowering operational costs, minimizing equipment failure due to power surges, and maximizing lifespan. This point-to-multipoint architecture allows a single optical fiber to serve multiple premises through splitters.

5.2 The FTTH Network Architecture Layers

The physical infrastructure of an FTTH network is divided into three distinct geographical segments:

A. The Feeder Segment: Connects the Central Office (CO) to the Fiber Distribution Hub (FDH). It uses heavy, high-fiber-count cables (48 to 144+ cores) to carry signals over long trunk routes (up to 20 km) without splitting.

B. The Distribution Segment: Spans from the FDH to street-level boxes like the Optical Distribution Point (ODP). Feeder fibers are split at the FDH, and distribution cables (12 to 24 cores) run along streets on poles or in ducts.

C. The Drop Segment: The final stretch running from the ODP to the customer’s ONT. It uses a single or dual-core drop cable designed with high bend tolerance (G.657 standard) to navigate tight corners inside home conduits.

5.3 Core Hardware Components

An FTTH network relies on a strict handshake between active hardware (requiring power) and passive hardware (no power):

5.3.1 Active Components (Power Required)

OLT (Optical Line Terminal): The central switch situated at the Central Office. It translates backbone internet traffic into optical pulses and coordinates downstream/upstream transmission scheduling.

ONT / ONU (Optical Network Terminal / Unit): The subscriber-end modem. It converts the incoming light signals from the drop cable into standard electrical Ethernet or Wi-Fi signals for household use.

5.3.2 Passive Components (No Power Required)

Optical Splitters: Passive glass components that divide a single input light wave into multiple output paths (e.g., 1:8, 1:16, 1:32) to feed multiple subscribers without power.

FDH / ODF Cabinets: Weatherproof enclosures hosting splitters and splice trays, securing connections against dust, water, and heat.

Chapter 6

FTTH Deployment, Coding and Testing

6.1 Fiber Color Coding Standard (TIA-598-C)

To safely identify individual fibers inside a multi-core cable, technicians use the industry-standard 12-color coding sequence. For cables with more than 12 fibers, the sequence repeats with a physical black tracer/stripe.

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	Blue	BL	7	Red	RD
2	Orange	OR	8	Black	BK
3	Green	GR	9	Yellow	YL
4	Brown	BR	10	Violet	VI
5	Slate (Grey)	SL	11	Rose (Pink)	RS
6	White	WH	12	Aqua	AQ

Table 6.1: TIA-598-C Color Coding Identification

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

Before installation, technicians perform a comprehensive site survey consisting of several phases:

- **Route Mapping:** Planning the shortest cable path from the street ODP to the customer premises.
- **Structural Assessment:** Finding entry points and deciding on the best indoor ONT location.
- **Safety Check:** Checking for power lines to maintain safe spacing and avoid electrical hazards.
- **BOM Calculation:** Creating the Bill of Materials including clamps, drop cables, and wall outlets.

6.2.2 Outdoor Routing Operations

Aerial Deployment: Drop wires are suspended between poles using drop clamps. Technicians leave slack loops to accommodate expansion and contraction from temperature changes.

Underground Deployment: Fiber is pulled or blown through buried HDPE (High-Density Polyethylene) conduits for maximum protection against environmental hazards.

6.2.3 Indoor Termination

The fiber cable enters the home and is terminated at a wall-mounted fiber outlet or rosette box. A highly flexible, bend-insensitive patch cord (typically G.657 standard) connects the ATB to the ONT, preventing signal loss around tight corners.

6.3 Splicing Methods Comparison

Feature	Fusion Splicing	Mechanical / Fast Connectors
Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

Table 6.2: Comparison of Fusion and Mechanical Splicing

6.4 Link Budget & Testing Benchmarks

FTTH operates over single-mode fiber using separate bidirectional wavelengths to avoid interference:

6.4.1 Wavelength Multiplexing

- 1490 nm: Carries downstream internet and voice traffic from OLT to subscriber ONT.
- 1310 nm: Used for upstream traffic uploads from ONT to the central OLT.
- 1550 nm: Allocated for downstream analog or digital TV overlays.

6.4.2 Testing Parameters

- Optical Power Meter (OPM): Used to measure the optical power level at the ONT. The normal range is -15 dBm to -25 dBm. Levels below -27 dBm indicate an attenuation fault.
- OTDR (Optical Time Domain Reflectometer): Sends light pulses down the fiber to analyze backscatter, mapping splice points and finding the exact location of cable cuts.

6.5 Troubleshooting & Indicators

Field diagnostics rely heavily on the ONT hardware status LED combinations:

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

Chapter 7

Field Visit: BSNL Exchange Office, Manjeri

7.1 Introduction to the Field Visit

As part of the internship in optical fiber communication, a technical visit was conducted to the BSNL Exchange office, Manjeri to gain practical exposure to the operation and management of fiber-optic communication networks. The visit provided an opportunity to observe how telecommunication services are delivered from the exchange to end users through optical fiber infrastructure.

7.2 Observations and Core Infrastructure

During the visit, we observed different equipment used in the exchange, including the Optical Line Terminal (OLT), fiber distribution panels, and other networking devices. The officials explained how internet, voice, and other communication services are transmitted through optical fiber cables from the exchange to customers. Behind the OLT, several backend devices are used for network management. The BNG (Broadband Network Gateway) handles user authentication, IP address allocation, routing, and internet session management. The MNGN (Multiplay Converged packet access Networks) acts as the packet transport layer, carrying FTTH data, voice, and business leased lines over a high-capacity ring network. The OLT connects to customer ONTs via the optical distribution network.

7.3 Conclusion of the Visit

The visit provided practical knowledge about the working of a telecommunication exchange and helped us relate the theoretical concepts of optical fiber communication to real-world applications. It was a valuable learning experience that improved our understanding of modern fiber-optic communication systems.

Part Conclusion: Fiber Optic Infrastructure

In conclusion, this module has successfully detailed the comprehensive operational framework of modern high-speed telecommunications. Beginning with the core waveguiding physics of single-mode fibers, the report systematically evaluated how light pulses are propagated, attenuated, and structured within field-grade multi-core distribution cables. The rigorous analysis of multi-stage passive optical splitters and mathematical power link budgeting demonstrated how a network ensures operational durability while supporting thousands of subscribers. Training with Bfone Fibernet provided hands-on experience in fusion splicing, fiber color coding, and diagnosing faults with OTDRs. Additionally, the BSNL Exchange visit illustrated macro-scale network architecture, showing the interaction between OLTs, BNG routers, and ONTs. Ultimately, we learned that successful FTTH networks rely on a balance of mathematical design and careful field installation to deliver reliable gigabit speeds.

Part II

Electrical Home Appliances Maintenance and Troubleshooting

Part Introduction: Domestic Electrical Engineering

The integration of specialized electro-mechanical hardware remains a vital pillar of the modern domestic environment. From electrothermal conversion units to complex induction power inverters and single-phase electromagnetic distribution motors, these household systems represent direct physical applications of fundamental thermodynamic and electromagnetic theories. This second module of the report provides a rigorous technical breakdown of consumer appliance electronics, diagnostic logic, and preventive maintenance protocols executed during the industrial training. Our objective was to understand the physics of appliance operation and safety systems rather than just replacing parts. The following chapters cover domestic wiring, series and parallel circuits, and safety devices like grounding, MCBs, and RCCBs that protect against shocks and short circuits. Next, we provide component analyses and troubleshooting steps for seven home appliances: the iron, induction stove, mixer, geyser, ceiling fan, wall fan, and water pump. This section concludes with practical calculations on household electricity auditing and charge transport.

Chapter 8

Fundamentals of Electrical Systems & Safety

8.1 Introduction

The modern domestic environment relies entirely on the seamless operation of electrical appliances. From food preparation to climate control and water management, these devices have transitioned from luxuries to everyday essentials. However, continuous operation inevitably leads to mechanical wear, insulation degradation, and electrical faults. This internship was undertaken to bridge the gap between abstract electrical theory and hands-on diagnostic engineering. The training structured under this module focused on developing a core competency in domestic wiring systems, understanding electrical safety protocols, and mastering the systematic teardown, repair, and preventive maintenance of seven foundational household appliances. The primary objective was not merely to fix broken devices, but to understand the physics governing their operation—such as electrothermal conversion, electromagnetism, and fluid dynamics—while strictly adhering to industrial safety standards.

8.2 Fundamentals of Electrical Systems & Safety Overview

Before working on any appliance, a solid foundation in circuit behavior and safety infrastructure was established.

8.2.1 Circuit Behaviors & Laws

- Ohm's Law & Power: We use $V = IR$ and $P = VI$ to calculate normal resistance values, allowing us to find short circuits and open windings during testing.
- Series Wiring: Current is constant while voltage drops. Used for safety switches, fuses, and thermostats to interrupt current flow.
- Parallel Wiring: Voltage remains constant while current divides. Standard for home wiring to ensure appliances operate independently.

$$V = IR \quad \text{and} \quad P = VI = I^2R \quad (8.1)$$

8.2.2 Earthing and Protection Systems

- Earthing: Connects the metal body of an appliance to the ground. If a hot wire touches the chassis, current flows to earth, preventing shock to the user.

- Fuses & MCBs: Devices that open the circuit during short circuits or overloads, protecting wiring from overheating.
- RCCB / ELCB: Detects small leakages (30mA) to ground by comparing live and neutral currents, tripping instantly to prevent severe shock.

Chapter 9

Technical Breakdown & Diagnostics of Core Home Appliances

9.1 Electric Iron

A. Working Principle

An electric iron works on the principle of the Joule's Heating Effect of Electric Current. When an electric current passes through a high-resistance conductor (heating element), it generates thermal energy ($H = I^2 R t$). Temperature regulation is achieved via a bimetallic strip thermostat. The strip consists of two different metals bonded together with distinct coefficients of thermal expansion. As the iron heats up, one metal expands faster than the other, causing the strip to bend and open the electrical contacts, breaking the circuit. Once it cools down, it straightens, closes the contact, and heat generation resumes.

$$H = I^2 R t \quad (9.1)$$

B. Key Components and Parts

- Soleplate: A smooth, heavy metal plate (aluminum/cast iron) that transfers heat to the fabric.
- Heating Element: A nichrome wire ribbon wrapped in mica sheets to insulate the live voltage while conducting heat.
- Bimetallic Thermostat: The temperature regulator with an adjustable dial that sets the contact gap.
- Thermal Fuse: A safety fuse that melts permanently if temperatures exceed safe limits due to thermostat failure.

C. Appliance Conclusion

Through the study of the electric iron, the practical application of electrothermal conversion and thermal feedback loops via bimetallic components was fully understood. Testing this device highlighted how crucial thermal fuses are in preventing catastrophic overheating.

9.2 Induction Cooker

A. Working Principle

Unlike traditional cooking methods that use thermal conduction, an induction cooker operates on the principles of Electromagnetism and Eddy Currents. The appliance takes 50Hz AC power

and converts it into a high-frequency (20kHz to 40kHz) alternating current using an internal inverter circuit. This high-frequency current passes through a copper induction coil beneath the ceramic plate, creating a dynamic, rapidly alternating magnetic field. When a ferromagnetic vessel is placed on top, this magnetic field induces circulating eddy currents within the base of the pan. Because the pan has electrical resistance, these currents generate heat directly inside the cookware itself, keeping the stovetop relatively cool. **B. Key Components and Parts**

- Induction Coil: A flat, spiral copper coil that creates the alternating magnetic field.
- IGBT: A high-power semiconductor switch that drives the high-frequency inverter circuit.
- Main PCB & Control Unit: The controller that processes settings, manages power level switching, and monitors safety sensors.
- Fan & Temperature Sensor (RTD): A cooling fan and thermistor that prevent overheating and detect boil-dry conditions.

C. Appliance Conclusion

The induction cooker diagnostics provided essential insights into solid-state electronics and high-frequency power switching. It demonstrated how magnetic fields can be manipulated for localized heating, emphasizing the need for robust PCB-level troubleshooting.

9.3 Mixer Grinder

A. Working Principle

The mixer grinder operates on electromagnetic mechanical conversion using an AC Universal Motor. A universal motor is a series-wound motor designed to run on either AC or DC power. When voltage is applied, current flows through both the stator (field windings) and the rotor (armature windings) via carbon brushes. This creates opposing magnetic fields, producing a high starting torque and high rotational speeds (up to 20,000 RPM), which are necessary for chopping, blending, and grinding food materials. **B. Key Components and Parts**

- Stator & Armature: The stationary field coils and rotating core that convert electrical energy to torque.
- Commutator & Brushes: Sliding carbon contacts that deliver current to the rotating commutator bars.

9.3.1 Ceiling Fan

A. Working Principle

A ceiling fan operates on the principle of a Single-Phase Induction Motor (Permanent Split Capacitor Motor). Single-phase AC power cannot generate a rotating magnetic field on its own to start a motor. To solve this, the motor has two separate windings: a Starting Winding and a Running Winding. A capacitor is wired in series with the starting winding to create a 90° electrical phase shift

in the current. This phase shift creates a simulated rotating magnetic field, which induces a current in the squirrel-cage rotor, generating the torque needed to turn the fan blades. **B. Key Components and Parts**

- Stator & Rotor: An inner stationary core with copper windings and an outer rotating steel shell.
- Run Capacitor: A 2.25 to 2.5 microfarad capacitor that creates the phase shift to start the motor.
- Ball Bearings: Dual bearings that support the rotor shell, minimizing noise and friction.
- Down-Rod: The steel pipe assembly that suspends the fan safely from the ceiling hanger.

C. Appliance Conclusion

Working with ceiling fans clarified the theory of phase splitting in single-phase induction systems. The practical experience emphasized the diagnostic process of isolating electrical winding failures from mechanical bearing resistance.

9.3.2 Wall Fan

A. Working Principle

A wall fan operates similarly to a ceiling fan, using a Single-Phase Permanent Split Capacitor Induction Motor, but includes specialized mechanisms for targeted directional airflow and oscillation. While the motor creates mechanical rotation using a starting winding, running winding, and a capacitor to shift the current phase, it also drives a mechanical synchronous link gear system or an independent secondary micro-motor to achieve horizontal oscillation (swiveling back and forth). Speed regulation is managed either via a multi-tapped transformer winding or an electronic TRIAC circuit on a PCB, controlling voltage delivery to the motor. **B. Key Components and Parts**

- Blades & Grille: Curved plastic/metal blades protected by a wire mesh safety guard.
- Oscillation Gearbox: A reduction gear assembly that converts high-speed motor rotation into a slow swiveling motion.
- Speed Selector: A mechanical rotary switch, pull-cord, or electronic circuit that adjusts voltage to regulate speed.
- Thermal Cut-Off (TCO): A thermal fuse inside the stator coils that cuts power if the motor overheats due to a jam.

C. Appliance Conclusion

The analysis of the wall fan highlighted how structural mechanisms (linkage gears) integrate with electrical induction motors. It offered an excellent case study on thermal protection embedded within motor windings and the diagnostic tracking of mechanical oscillation failures.

Chapter 10

Troubleshooting & Maintenance Practice Matrix

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 10.1: ONT Troubleshooting Matrix Summary

Appliance	Common Symptom	Root Cause Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord	Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT	Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Overload switch tripped	Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element	Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor	Desoldered/disconnected old capacitor, verified value, and installed new $2.5\mu\text{F}$ unit.
Wall Fan	Fan rotates but does not oscillate / Overheating	Stripped linkage gears / Dry rotor shaft	Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor	Freed impeller manually; replaced bearings and capacitor.

Table 10.2: Appliance Troubleshooting & Maintenance Practice Matrix

10.1 Internship Outcomes & Skills Acquired

- **Safety First Mentality:** Learned to always check for live voltage using a test pen, work on rubber mats, and verify ground lines before powering up devices.
- **Structured Troubleshooting:** Mastered the process of tracing faults along the circuit path, starting from the plug, checking fuses, and then testing switches and windings.
- **Multimeter Proficiency:** Gained hands-on experience using multimeters to test continuity, winding resistances, and checking live AC voltages safely.

Chapter 11

Applied Practical Calculations

11.1 Domestic Electricity Consumption Auditing

To calculate household power consumption, energy is tracked in kilowatt-hours (kWh), where 1 Unit = 1 kWh.

Determine the exact 30-day billable consumption for a household running two 40W tube lights for 6 hours daily, three 65W fans for 8 hours daily, and separate appliances drawing 150W for 2 hours daily.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = 70.2 \text{ kWh (Units)}$$

11.2 Charge Calculation in Audio Systems

A 230V system consumes a total of 8 MJ of energy during operation. Determine the absolute electrical charge that moved through the circuit.

$$\text{Given: } V = 230 \text{ V, } E = 8 \times 10^6 \text{ J}$$

$$\text{Equation: } E = qV$$

$$\begin{aligned} q &= \frac{E}{V} \\ &= \frac{8 \times 10^6 \text{ J}}{230 \text{ V}} \\ &= 3.48 \times 10^4 \text{ Coulombs (C)} \end{aligned}$$

Part Conclusion: Domestic Electrical Engineering

In conclusion, this second module has established a robust technical bridge between undergraduate physics and field-level electrical repair logic. By systematically mapping out internal component frameworks across seven fundamental consumer appliances, the training provided deep insights into everyday electrical systems. Tracing diagnostic signatures—such as a bimetallic contact opening in a thermal loop or a split-phase capacitor drop in an induction motor—showed how equipment behaviors can be accurately tracked using standard multi-meter analysis. Hands-on practice showed how safety elements like thermal fuses (TCOs) and overload switches (OLPs) prevent fires and protect equipment. Furthermore, calculating energy consumption and charge transfer helped us understand power management. Ultimately, the troubleshooting methods and repair skills learned in this course provide a valuable practical toolkit for maintaining home appliances safely.